Fault Code: 339 (Integrated)

Idle Shutdown Vehicle Accessories Relay Driver Circuit - Voltage Below Normal or Shorted to Low Source

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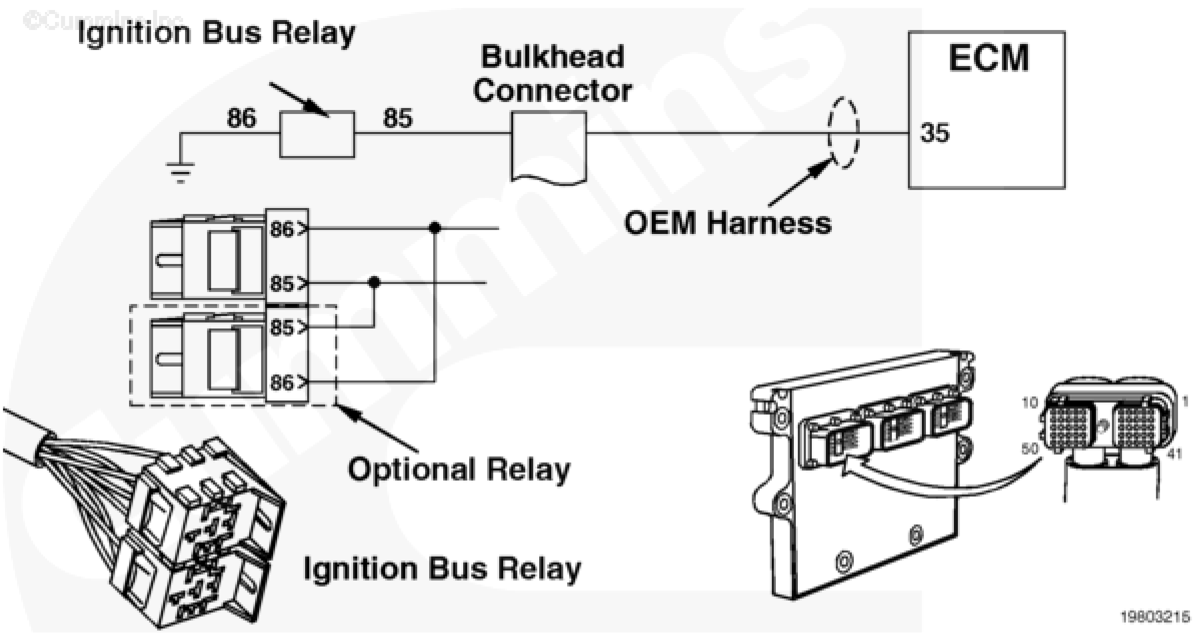
Overview

Codes	Reason	Effect
Fault Code: 339 PID(P): S087, 4 SPN: 1267 FMI: 4 Lamp: Yellow SRT:	Idle Shutdown Vehicle Accessories Relay Driver Circuit - Voltage Below Normal or Shorted to Low Source. Less than 6 VDC detected at the ignition bus relay output circuit when the high voltage was expected by the engine electronic control module (ECM).	The ICON™ system will be disabled. **Only** mandatory shutdown will be enabled. Constant power will be at the keyswitch ignition circuit.



Circuit Description

The ignition bus relay controls ignition circuits powering the heating and air conditioning controls and other equipment connected to the ignition bus relay(s). This relay(s) is controlled by ignition relay positive (+) signal from the engine ECM OEM connector pin 35.

Component Location

The ignition bus relay is located under the dash inside the vehicle cab.

Shoptalk

This fault typically indicates a short circuit to ground or an open circuit from the engine ECM connector pin 35, ignition relay positive (+). Ignition relay positive (+) (pin 35) outputs 12 VDC to open the ignition bus relay(s) when the ICON™ system has powered down the vehicle and needs to disconnect power going to the cab circuit. The ICON™ bus relay(s) is normally closed when no power is applied.

Warnings and Cautions

⚠ CAUTION ⚠

To reduce the possibility of damaging a new engine ECM, all other active fault codes must be investigated prior to replacing the engine ECM.

To reduce the possibility of pin and harness damage, use the following test leads when taking a measurement:
Part Number 3822917 - female Deutsch/AMP/Metri-Pack test lead
Part Number 3822758 - male Deutsch/AMP/Metri-Pack test lead.

Troubleshooting Summary

STEPS		**SPECIFICATIONS**
STEP 1.	Read all fault codes.	
	STEP 1A. Read fault codes with INSITE™ electronic service tool or flash out with ICON™ lamp.	Fault Code 339 inactive
STEP 2.	Check the ignition bus relay.	
	STEP 2A. Perform the ignition bus relay test with INSITE™ electronic service tool.	Dashboard blowers turn off

STEPS		SPECIFICATIONS
STEP 3.	Inspect the OEM harness connector and the fire wall bulkhead OEM harness connector.	
	STEP 3A. Inspect the OEM harness and the engine ECM connector pins.	No damaged pins
	STEP 3B. Check for a short circuit to ground.	More than 100k ohms
	STEP 3C. Check for a short circuit from pin to pin.	More than 100k ohms
	STEP 3D. Check for an open circuit.	Refer to the OEM troubleshooting and repair manual for specifications
STEP 4.	Check the ignition bus relay.	
	STEP 4A. Check the ignition bus relay connector pins.	No damaged pins
	STEP 4B. Check the ignition bus relay coil for a short circuit.	Refer to the OEM troubleshooting and repair manual for specifications
STEP 5.	Clear the fault code.	
	STEP 5A. Disable the fault code.	Fault Code 339 inactive; Dashboard blowers turned off

STEP 1. Read all fault codes.

STEP 1A. Read fault codes with INSITE™ electronic service tool or flash out with ICON™ lamp.

Conditions :
- Turn keyswitch ON.

Action	Specification/Repair	Next Step
• Read the fault codes using INSITE™ electronic service tool or flash out ICON™ lamp.	Fault Code 339 inactive	**2A**
	Fault Code 339 active	**2A**

STEP 2. Check the ignition bus relay.

STEP 2A. Perform the ignition bus relay test with INSITE™ electronic service tool.

Conditions :

- Turn keyswitch ON.

Action	Specification/Repair	Next Step
• Refer to Procedure 019-305 (/qs3/pubsys2/xml/en/procedures/97/97-019-305.html).	Dashboard blowers turn off	**5A**
	Does **not** meet specifications	**3A**

STEP 3. Inspect the OEM harness connector and the fire wall bulkhead OEM harness connector.

STEP 3A. Inspect the OEM harness and the engine ECM connector pins.

Conditions :

- Turn keyswitch OFF.
- Disconnect the OEM harness connector from the engine ECM and the OEM harness bulkhead connector.

Action	Specification/Repair	Next Step

• Corroded pins • Bent or broken pins • Pushed back or expanded pins • Wire insulation damage • Moisture in or on the connector • Missing or damaged connector seals • Connector shell broken • Dirt or debris in or on the connector pins. For general inspection techniques, refer to Component Connector and Pin Inspection, Procedure 019-361 (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html).	No damaged pins	**3B**

	Repair the damaged pins. • Flush the dirt, debris, and moisture from the connector pins using electrical contact cleaner, Part Number 3824510. • Install the appropriate connector seal if it is damaged or missing. Repair or replace the OEM wiring harness. • Refer to the OEM service manual. Replace the engine ECM. • Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/79/79-019-031.html) in Troubleshooting and Repair Manual, CELECT™ Plus, Bulletin 3666130, or • Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/82/82-019-031.html) in Troubleshooting and Repair Manual, Electronic Control System, ISM, Bulletin 3666266, or • Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/72/72-019-031.html) in Troubleshooting and Repair Manual, Electronic Control System, Signature and	**5A**

	ISX, Bulletin 3666259, or • Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/88/88-019-031.html) in Troubleshooting and Repair Manual, Electronic Control System, CM870 ISM, Bulletin 4021381, or • Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/70/70-019-031.html) in Troubleshooting and Repair Manual, Electronic Control System, CM870 Signature and ISX, Bulletin 4021334, or • Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/12/12-019-031.html) in Troubleshooting and Repair Manual, Electronic Control System, CM875 ISM, Bulletin 4021477.	

STEP 3B. Check for a short circuit to ground.

Conditions :

- Turn keyswitch OFF.
- Disconnect the ignition bus relay from the OEM harness.
- Disconnect the OEM harness connector from the engine ECM.

Action	Specification/Repair	Next Step

• Measure the resistance from pin 35 (or idle shutdown) of the OEM harness connector to the engine block ground. Refer to the wiring diagram or the circuit description at the beginning of this fault code for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	More than 100k ohms	**3C**
	Repair or replace the OEM wiring harness. • Refer to the OEM service manual.	**5A**

STEP 3C. Check for a short circuit from pin to pin.

Conditions :

- Turn keyswitch OFF.
- Disconnect the OEM harness connector from the engine ECM.
- Disconnect the ignition bus relay(s) from the OEM harness.

Action	Specification/Repair	Next Step

• Measure the resistance from pin 35 to all other pins in the OEM harness connector.	More than 100k ohms	**3D**
Refer to the wiring diagram or the circuit description at the beginning of this fault code for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	Repair or replace the OEM wiring harness. • Refer to the OEM service manual.	**5A**

STEP 3D. Check for an open circuit.

Conditions :

- Turn keyswitch OFF.
- Disconnect the OEM harness connector from the engine ECM.
- Disconnect the OEM harness from the ignition bus relay.

Action	Specification/Repair	Next Step

• Measure the resistance from the ignition bus relay connector pin 85 to the OEM harness connector at engine ECM pin 35. Refer to the wiring diagram or the circuit description at the beginning of this fault code for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	Refer to the OEM troubleshooting and repair manual for specifications.	**4A**
	Repair or replace the OEM wiring harness. • Refer to the OEM service manual.	**5A**

STEP 4. Check the ignition bus relay.

STEP 4A. Check the ignition bus relay connector pins.

Conditions :

- Turn keyswitch OFF.
- Disconnect the OEM harness from the ignition bus relay.

Action	Specification/Repair	Next Step

• Corroded pins • Bent or broken pins • Pushed back or expanded pins • Wire insulation damage • Moisture in or on the connector • Missing or damaged connector seals • Connector shell broken • Dirt or debris in or on the connector pins. For general inspection techniques, refer to Component Connector and Pin Inspection, Procedure 019-361 (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html).	No damaged pins	**4B**

	Repair the damaged pins. • Flush the dirt, debris, and moisture from the connector pins using electrical contact cleaner, Part Number 3824510. • Install the appropriate connector seal if it is damaged or missing. Repair or replace the OEM wiring harness. • Refer to the OEM service manual. Replace the engine ECM. • Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/79/79-019-031.html) in Troubleshooting and Repair Manual, CELECT™ Plus, Bulletin 3666130, or • Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/82/82-019-031.html) in Troubleshooting and Repair Manual, Electronic Control System, ISM, Bulletin 3666266, or • Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/72/72-019-031.html) in Troubleshooting and Repair Manual, Electronic Control System, Signature and	**5A**	

	ISX, Bulletin 3666259, or • Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/88/88-019-031.html) in Troubleshooting and Repair Manual, Electronic Control System, CM870 ISM, Bulletin 4021381, or • Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/70/70-019-031.html) in Troubleshooting and Repair Manual, Electronic Control System, CM870 Signature and ISX, Bulletin 4021334, or • Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/12/12-019-031.html) in Troubleshooting and Repair Manual, Electronic Control System, CM875 ISM, Bulletin 4021477.	

STEP 4B. Check the ignition bus relay coil for a short circuit.

Conditions :

- Turn keyswitch OFF.
- Disconnect the ignition bus relay from the OEM harness.

Action	Specification/Repair	Next Step

• Measure the resistance from pin 86 of the ignition bus relay(s) to pins 30, 87, and 87A of the relay(s). Refer to the wiring diagram or the circuit description at the beginning of this fault code for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	Refer to the OEM troubleshooting and repair manual for specifications	**5A**
	Replace the idle shutdown vehicle accessory or ignition bus relay, refer to the OEM troubleshooting and repair manual.	**5A**

STEP 5. Clear the fault code.

STEP 5A. Disable the fault code.

Conditions :

- Connect all components.
- Turn keyswitch ON.

Action	Specification/Repair	Next Step

- Verify Fault Code 339 is inactive using INSITE™ electronic service tool. - Perform ignition bus relay test using INSITE™ electronic service tool. - Erase the inactive fault codes using INSITE™ electronic service tool.	Fault Code 339 inactive; Dashboard blowers turned off	**Repair complete**
	Return to the troubleshooting steps, or contact the nearest Cummins Authorized Repair Location if all the steps have been completed and rechecked. Troubleshoot any remaining active fault codes.	**Appropriate troubleshooting charts**

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